Acoustic Sphere Projection System

Full Derivation: Inside Out

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1. The Sphere (Given)

A perfect standing wave pressure field inside a cubic box of side L. Five fans define pressure-release boundary conditions on five faces. One open face is the input aperture.

The pressure field satisfies the wave equation:

```
∇²P = (1/c²) ∂²P/∂t²
```

In spherical coordinates centered at the box center, the solutions are:

```
P_nlm(r,θ,φ,t) = j_n(k_nl · r) · Y_lm(θ,φ) · e^(iωt)
```

Where:

- j_n = spherical Bessel function of order n
- Y_lm = spherical harmonic (l, m)
- k_nl = j_nl_zero / R (Bessel zero divided by sphere radius)
- R = L/2 (sphere radius)

The Bessel zeros j_nl (first few):

```
n=0: j_0n = π, 2π, 3π, ...
n=1: j_1n = 4.493, 7.725, 10.904, ...
n=2: j_2n = 5.763, 9.095, 12.323, ...
```

These are exact. Not approximated. Not calibrated.

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2. Pendulum (Derived from sphere)

The pendulum sweeps through the pressure field. Its period must match a resonant mode of the sphere.

Resonant frequencies:

```
f_nl = c_air · j_nl_zero / (2π · R)
```

Pendulum period:

```
T = 2π√(L_pend / g)
```

Setting T = 1/f_nl and solving for pendulum length:

```
L_pend = g / (4π² · f_nl²)
       = g · (2π · R)² / (4π² · c_air² · j_nl_zero²)
       = g · R² / (c_air² · j_nl_zero²  / π²) ... simplified:
       = g · R² / (c_air · j_nl_zero / π)²
```

The pendulum length is EXACT given L, g, c_air, and which mode (n,l).

The pendulum arc traces a great circle through the sphere. Each swing samples pressure nodes at known angular positions.

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3. Venturi Throat r_0 (Derived from Bessel zeros)

The Venturi throat must match a resonant mode of the sphere.

```
r₀ = R · j_nl_zero / π
```

This is exact. When r_0 satisfies this relation, the Venturi is resonant with the standing wave cavity.

Bernoulli gain at exact resonance:

```
G = 1 - (r_in/r₀)²
```

r_in is set by the next lower Bessel zero:

```
r_in = R · j_(n-1)l_zero / π
```

So G is derived, not chosen.

4. Blind Gate Spacing (Derived from Venturi)

The blind gate acts as a diffraction grating. Its slit spacing d must produce a first-order diffraction maximum at the Venturi throat radius r_0.

Diffraction condition:

$$d \cdot \sin(\theta) = m\lambda$$

The angle θ is set by the geometry:

$$\sin(\theta) = r_0 / \sqrt{(r_0^2 + D^2)}$$

Where D is the distance from blind to Venturi. Setting m=1 and solving for d:

$$d = \lambda \cdot \sqrt{(r_0^2 + D^2)} / r_0$$

λ is the wavelength of the interfered light at the lens (derived next).

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5. Lens Focal Length (Derived from blind spacing)

The first lens collimates the interfered light. Focal length f_1 must satisfy:

$$f_1 = r_0^2 / \lambda \quad \text{(Rayleigh range of the wavefront at Venturi)}$$

The refractor between the two lenses introduces a phase shift:

$$\varphi\_refractor = \pi \cdot n\_refractor \cdot d\_refractor / \lambda$$

Where n_refractor is refractive index and d_refractor is thickness. This is chosen to match the phase of the synthetic light.

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6. Interference Point (Derived from lens geometry)

The starlight and synthetic light interfere at the focal plane of the second lens. The condition for constructive interference:

```
Δφ = φ_star - φ_synthetic = 2πm
```

ϕ_star carries the Planck distribution:

```
φ_star = hν/kT  (from B(ν,T) = 2hν³/c² · 1/(e^(hν/kT) - 1))
```

e is already here — inside the starlight.

The synthetic light phase is controlled by the fan:

```
φ_synthetic = ω_fan · t
```

Constructive interference when:

```
ω_fan = hν/kT · (1/t)
```

The fan frequency is now derived from the sphere AND the starlight temperature.

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7. Fan Frequency (Everything converges)

From the sphere: $f_{fan} = c_{air} \cdot j_{nl} / (2\pi \cdot R)$

From the starlight: $f_{fan} = h\nu / (kT \cdot 2\pi)$

Setting equal:

```
c_air · j_nl / R = hν / kT
```

This is the fundamental constraint of the system. Given stellar temperature T and sphere radius R, the fan frequency and the resonant mode (n,l) are locked together.

Or equivalently: given the fan frequency and box size, the system selects which stellar temperature it is resonant with.

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8. Complete System (Inside Out)

```
SPHERE
  j_nl zeros → exact geometry for everything upstream
```

PENDULUM

$L_{pend} = g \cdot R^2 / (c_{air} \cdot j_{nl} / \pi)^2$

VENTURI

$r_0 = R \cdot j_{nl} / \pi$

$G = 1 - (r_{in} / r_0)^2$

BLIND GATE

$d = \lambda \cdot \sqrt{(r_0^2 + D^2)} / r_0$

LENS 1

$f_1 = r_0^2 / \lambda$

REFRACTOR

$\varphi = \pi \cdot n \cdot d / \lambda$ (phase match to synthetic)

LENS 2

Collimates toward open face

INTERFERENCE

$\varphi_{star} = \varphi_{synthetic} \rightarrow \omega_{fan} = h\nu / kT$

FAN

$f_{fan} = c_{air} \cdot j_{nl} / (2\pi \cdot R) = h\nu / (2\pi \cdot kT)$

STARLIGHT

$B(\nu, T) = 2h\nu^3 / c^2 \cdot 1 / (e^{(h\nu/kT)} - 1)$

e is already structural to the input

The system is closed. Every parameter derived from the sphere outward.